

Progress Report: High-Throughput Monte Carlo Simulations and System Response Matrix Modeling for a Novel Brain SPECT Architecture

ACCESS allocation: MDE260019 – ACCESS Explore, 400,000 SU award (Slurm account mgh102, SDSC Expanse)
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Reporting period: 2026-08-14 (allocation start) – 2026-09-15
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1. Scientific objective

This allocation supports Monte Carlo simulation of a novel, fixed-gantry brain-dedicated SPECT scanner (DB-SPECT) originally proposed by our group [1] as an alternative to conventional dual-head gamma cameras and adapted multi-pinhole systems for epilepsy, dementia, Alzheimer's, Parkinson's, and tumor imaging. The initial design concept [1] used 118 large-area detector modules on a spherical gantry viewing a 210 mm-diameter field of view through one-collimator-per-detector optics, and demonstrated (via analytical modeling and GATE Monte Carlo) up to 18.3x the sensitivity of a dual-head gamma camera at matched resolution, and 4.05 mm FWHM resolution at matched sensitivity. The current campaign simulates a refined 73-head iteration of this architecture (625 pixels/head) in GATE 10 (OpenGATE) to produce high-statistics system response matrices (SRMs) – the sensitivity/detection-probability mapping between image-space voxels and detector pixels – at three spatial resolutions (1, 1.5, 2 mm). These SRMs are the forward-model operator required for quantitative iterative image reconstruction (e.g. OSEM) and for evaluating the updated design's reconstructed-image sensitivity and resolution ahead of hardware fabrication. Producing them to the statistical precision needed for a 210 mm-diameter uniform source over a ± 105 mm reconstruction volume requires simulating on the order of 4×10^{13} primary photon histories, which is only tractable with a large-scale allocation such as this one.

2. Computational approach

Each SRM voxel-pixel entry is accumulated from single-photon Monte Carlo histories tracked through the scanner geometry (collimators, crystal stack, shielding) with GATE 10's Geant4 backend. The workload is embarrassingly parallel across independent primaries, and is decomposed as follows:

- The full campaign requires 4.0005×10^{13} simulated primaries, split into **8 sequential parts** of 5.000625×10^{12} primaries each.
- Each part is a 126-way SLURM job array on Expanse's shared partition, 127 CPUs per task (whole-node), 20-hour wall-clock limit per task.
- Each array task simulates 5 independent loops of 7.9375×10^9 primaries, converts each loop's raw single events into a sparse (coordinate, count) representation immediately (sparse SRM accumulation), then combines its own 5 loops into a per-task partial SRM.
- Per-task partial SRMs are pulled to the home workstation over Globus and incrementally combined into per-part, then campaign-level, sparse SRMs, split into one compressed CSR matrix per detector head for efficient downstream reconstruction I/O.

3. Progress to date

As of this report, the 8-part brain-SPECT campaign (group brain_40t_20260910T020837Z, started 2026-09-09) stands at:

Metric	Value
Target primaries (full campaign)	4.0005×10^{13}
Committed primaries	1.2422×10^{13} (31.05 %)
Committed + in-flight primaries	1.2573×10^{13} (31.43 %)
Array tasks complete	313 / 1,008 (31.05 %)
Simulation loops complete	1,565 / 5,040 (31.05 %)

Part	Tasks complete	Committed primaries	In-flight primaries	Status
1	123 / 126	4,881,560,589,347	31,750,070,087	Nearly done; 3 straggler tasks
2	126 / 126	5,000,625,642,007	0	Complete
3	64 / 126	2,539,997,583,860	119,062,730,888	In progress
4–8	0 / 126 each	0	0	Not yet submitted

Part 2 is fully complete and part 1 is 98% complete, validating the end-to-end pipeline (simulation, sparse SRM extraction, per-task combine, Globus transfer, workstation aggregation) at production scale across multiple independent job arrays. Part 3 is roughly half complete and actively running. Parts 4–8 (60% of the total campaign) remain to be submitted.

4. Resource utilization

Based on the 313 completed array tasks reported above (each requesting 127 CPUs for up to 20 hours):

Metric	Value
Core-hours consumed to date (estimated)	$\approx 682,400$ core hours
Mean core-hours per completed task	$\approx 2,180$ core hours
Projected core-hours, full 40,005-primary campaign	≈ 2.20 M core hours
Projected core-hours remaining (695 tasks)	≈ 1.52 M core hours
Current ACCESS Explore award	400,000 core hours

The consumed and projected figures above are our own extrapolation from measured per-task wall-clock time on completed tasks (127 requested CPUs $\times$ task wall-clock seconds), **not** a pull from the ACCESS/XSEDE usage ledger; the PI should confirm actual usage-to-date against the ACCESS allocations portal before this report is finalized. Even accounting for estimation error, the projected total need (≈ 2.2 M core hours) is roughly 5.5x the 400,000-SU Explore award, so this campaign cannot be completed on an Explore-tier allocation and requires a resource increase. These projections will be refined as parts 4–8 proceed. Storage on Expanse is used only as a rolling buffer (per-task outputs are under 1 GB and are pulled to the PI's workstation and purged within days of completion), so no persistent-storage allocation growth is anticipated.

5. Technical issues identified and resolved

Two defects were found during progress review this period and corrected in the pipeline:

1. **Lost statistics from straggler tasks.** Array tasks that reach a SLURM time limit mid-campaign (3 of 126 in part 1; more expected in later parts) previously contributed zero data to the SRM, even when several of their 5 loops had fully completed and been written to disk – their task-level combine step never ran. A repair utility now recovers these completed loops into a valid partial per-task SRM instead of discarding them.

2. **Missing primaries denominator in SRM metadata.** The per-task combine step was not recording how many primaries were behind each SRM (`simulated_primaries` was stuck at 0), which silently breaks the sensitivity calculation ($\text{sensitivity} = \frac{\text{accumulated_counts}}{\text{simulated_primaries}}$). The simulation wrapper now computes this value from each task’s own simulation-statistics files and records it correctly; a repair script patches already-produced metadata without altering the underlying SRM data.
3. **Dashboard progress percentage.** The live monitoring dashboard’s multi-part aggregation only counted parts that had already produced a progress report, so it displayed 82.8% complete instead of the correct 31.05%. This has been corrected to count all 8 manifested parts.

None of these affected already-published or already-transferred SRM data beyond the metadata field noted above; all fixes are additive/corrective and have been validated against the current campaign’s mounted output.

6. Remaining work and timeline

- Finalize the 3 straggler tasks in part 1 and continue monitoring part 3 to completion.
- Submit parts 4–8 (695 remaining array tasks, ≈ 1.52 M SU estimated), sequenced to stay within per-part Lustre buffer limits.
- Combine the completed parts into the campaign-level, per-head sparse SRMs and validate reconstructed-image sensitivity against the expected 84–100% acceptance range established during pipeline validation.
- Evaluate GPU-accelerated combine tooling on the workstation to shorten the aggregation step once multiple parts are combinable simultaneously.

7. Resource request

The current award (ACCESS Explore, 400,000 core hours) is insufficient to complete this campaign: our estimate of $\approx 682,400$ core hours already consumed for 31% of the campaign already exceeds the Explore cap, and the full 4.0005×10^{13} -primary brain-SPECT SRM is projected to require ≈ 2.2 million core hours in total (≈ 1.52 million core hours to complete the remaining 69%). We therefore request an increase beyond the Explore tier (e.g. an ACCESS Discover- or Maximize-tier award, or a supplement covering the shortfall) sized to the ≈ 1.52 million SU remaining need above, subject to confirming exact consumption-to-date against the ACCESS portal. No change to storage or other resource types is requested; compute is the only resource type consumed by this workload.

8. Conclusion

The brain-SPECT SRM production pipeline is validated at scale: two of eight campaign parts are complete or nearly complete, the transfer/aggregation pipeline has processed over 300 array tasks without data loss once the straggler-recovery fix is applied, and the remaining compute need is well-characterized from measured per-task cost. We request continued access to the awarded allocation to complete parts 4–8 and deliver the final campaign-level system response matrices.

Bibliography

- [1] Y. Feng, A. Hashemi, S. Soleymani, M. Ottensmeyer, and H. Sabet, “DB-SPECT, a Fixed-Gantry SPECT Scanner for Dynamic Brain Imaging: Design Concept and First Results,” in *2022 IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC)*, 2022, pp. 1–3. doi: [10.1109/NSS/MIC44845.2022.10399152](https://doi.org/10.1109/NSS/MIC44845.2022.10399152).